

Location-based Non-linear Playback of AR Task Recording

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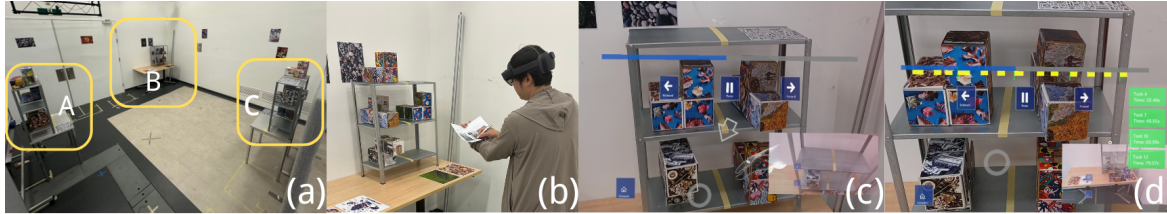


Figure 1: System overview. (a) A physical task environment where objects are moved and arranged between three task locations, each spaced 3 metres apart. (b) A user reviews an Augmented Reality task recording at location B. (c) With a conventional linear playback interface, participants need to manually navigate the timeline to find moments relevant to the current location. (d) With Location-based non-linear playback, the system automatically segments the recording based on spatial data and highlights the task segments on the timeline as yellow bars. A contextual task list panel (shown in green) displays the task segments associated with the user's current location, streamlining navigation and review.

ABSTRACT

Augmented Reality (AR) systems that capture and replay user actions provide valuable support for training and task guidance by preserving activities within their original spatial context. However, reviewing AR recordings remains inefficient, as most systems rely on linear playback that requires time-consuming manual search. This project investigates a non-linear replay approach that automatically segments AR task recordings based on spatial location data. The system enables reviewers to navigate recordings by jumping directly to task segments associated with their current physical location. Preliminary findings from a user evaluation suggest that organizing AR recordings around spatial structure can better support task review and analysis, and provide a foundation for more efficient and usable AR training and evaluation systems.

Index Terms: Location-Based Augmented Reality, Asynchronous Collaboration.

1 INTRODUCTION

Augmented Reality (AR) has been widely recognized as an effective tool for training and task guidance, offering real-time support and immersive interaction [6, 3, 16]. The AR task guidance systems that are capable of capturing and replaying user's actions are not only essential for trainees to revisit expert demonstrations but also useful for experts to review and assess the trainee's action and provide personalized feedback [1, 2]. As such, efficient performance evaluation directly supports iterative training by helping to identify mistakes, reinforce correct behaviors, and adapt future instructions. However, reviewing AR recordings for performance evaluation remains a time-consuming and inefficient process, especially when relying on conventional linear playback. In this case, experts tasked with assessing user performance must manually search through lengthy recordings to identify potential mistakes, making it challenging to extract critical insights. This manual review process not only increases task load but also hinders timely and precise feedback.

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This project aims to address these challenges by developing an automated system that segments task recordings based on the recorder's location data (see Figure 1). The system is designed to help experts efficiently review recorded task sessions, focusing on identifying errors and performance deviations. By using location-based segmentation, the system organizes recorded sessions spatially, allowing experts to navigate directly to key task regions where mistakes are likely to occur. This structured review approach reduces reliance on linear playback, enabling a more efficient and targeted analysis of user performance.

The key contribution of this project lies in the development of an intelligent segmentation platform that enhances expert review by structuring AR task recordings according to spatial context. By improving access to critical task moments that are relevant to the viewer's location, this system aims to enhance the overall efficiency and effectiveness of expert-driven AR-based task evaluation.

2 RELATED WORK

Automatic annotation in task guidance systems aims to improve instructional clarity and efficiency by dynamically generating contextual markers, highlights, or instructions based on user actions and environmental conditions. Prior research has primarily focused on various techniques for generating annotations, including automated data labeling for human activity recognition [4] and semantic extraction from 3D point cloud data [7]. Recent examples used Large Language Models (LLMs) for automated annotation [13], demonstrating the potential of AI-based annotation workflows. Yadav et al. [18] further explored how GPT-4 can automate text annotation, employing AI-driven techniques to enhance semantic proximity labeling. While these studies present diverse annotation strategies, they are primarily focused on generating annotations as stand-alone processes rather than integrating them into real-time AR task guidance systems.

To address this gap, several studies combined automatic annotation with AR-based task guidance to create more interactive, real-time instructional environments. Oomori et al. [11] presented a system that enhances video-based learning of physical skills by automatically generating flowcharts from instructional video transcripts using LLMs. Similarly, Bihari et al. [1] presented a novel AI-driven Mixed Reality (MR) training platform that automates the creation of step-by-step instructional materials for industrial training. Skreini et al. [15] introduced an immersive VR-based author-

ing tool for generating instructional content for industrial maintenance and assembly procedures. Their system enabled experts to demonstrate tasks in a VR environment, with automatic step segmentation and real-time recording of hand movements, tool usage, gaze, speech, and machine part interactions.

While significant progress has been made, there remains a critical disconnect between spatially contextualized task recordings and their effective utilization during their review. Most location-based AR studies focus on user-facing applications for navigation [14], education [11], or entertainment [12], often requiring manual location tagging or predefined marker setups, which limits scalability and efficiency. Conversely, advances in automatic annotation have largely prioritized stand-alone or offline processing [5, 17], with limited integration into dynamic, real-time task guidance or post-task review scenarios. Furthermore, prior AR training systems predominantly relied on linear playback for task recording, which is inherently time-consuming and cognitively demanding. Recently, researchers explored non-linear playback mechanisms based on object manipulation, allowing viewers to navigate recordings based on task context [2]. This gap represents a missed opportunity to streamline expert evaluations by aligning task segmentation with users' physical interactions in space.

Our work bridges this gap by introducing an automated, location-based segmentation framework tailored for a non-linear review of AR task recordings. By enabling intuitive navigation to spatially defined task segments, our system reduces review time and task load, representing a novel contribution to AR task analysis and expert performance evaluation.

3 SYSTEM DESIGN AND IMPLEMENTATION

As a proof of concept, this project developed a prototype system that automates the capture, segmentation, and playback of task sessions using spatial and interaction cues. This section outlines the design of the system including three components: (1) the task recording mechanism that captures spatial and contextual data during user activities, (2) the segmentation logic that identifies and structures segments from task recording based on user behavior and location, and (3) the recording review AR interface that enables intuitive and non-linear navigation through segmented content. Together, these components aim to streamline the review process, reduce task load, and improve the usability of AR task recording and replay systems that support training applications.

3.1 AR Task Recording

The AR task recording feature of the system was developed based on previous research [8, 10] and designed to enable users to capture and record their physical task activities using the Microsoft HoloLens 2 AR headset. The system automatically captures key contextual data of user's actions, including the user's location, head and hand poses, and gaze point at a sampling rate of 10Hz, along with a first-person-view video.

To enhance the flexibility and usefulness of the recorded content, users have the option to manually add annotations at their gaze point during their task performance. Upon completion of a recording session, the system systematically saves the anchor locations, contextual data, and the captured video. Subsequently, the system performs automatic segmentation of the recorded data based on predefined spatial and interaction criteria, transforming the recordings into structured segments. The specific segmentation methodology and its application criteria are elaborated on in the following section.

3.2 Segmentation of AR Task Recording

The system automates the segmentation of task recording by analyzing the recorder's spatial and interaction data, ensuring a structured and efficient documentation of the captured tasks. As de-

scribed earlier, the AR recording captures critical contextual information, including the recorder's location, head and hand poses, and focus point, along with a first-person-view video, providing a comprehensive record of task execution. To enable automatic segmentation, our system uses predefined spatial and gaze-based thresholds that reflect natural human movement patterns. Unlike prior object-centric methods that rely on computer vision techniques [9] that are better suited to stationary, single-scene tasks and often struggle in visually repetitive environments, our approach segments based on the user's physical location and gaze behavior. A task segment is defined when the recorder remains within a one-meter radius for more than one second while maintaining gaze within that same range, indicating focused activity in a localized zone. These parameters, derived from typical walking speed and arm reach, help capture meaningful task boundaries across multiple locations. To minimize fragmentation, consecutive segments are merged if separated by less than 1.5 seconds. This process structures raw AR recordings into spatially organized segments, supporting efficient non-linear review and reducing redundant navigation efforts.

The segmentation approach allows for a more intuitive and task-oriented review experience, eliminating the need to manually locate key moments within lengthy recordings. By automatically identifying and grouping task segments, the system improves the usability of AR task recording playback for training purposes, both task guidance and review, facilitating more efficient knowledge transfer and performance evaluation (see Figure 2).

3.3 Review of AR Task Recording

The system enhances the review process by providing a structured visualization of the segmented tasks on the timeline. Each task segment is distinctly highlighted in yellow on the timeline with its start and end times (see Figure 1(d)), allowing users to seamlessly navigate and jump directly to relevant moments within the AR task recording. This structured timeline aims to reduce the time required to review task recording compared to the conventional timeline-based playback interface.

In addition to timeline-based navigation, the system incorporates a location-based task retrieval mechanism (see Figure 2). If a reviewer is near a recorded task location, the system automatically displays all associated tasks at the viewer's current location on a task panel, as shown in Figure 1(d). This feature enables users to quickly access relevant segments of AR recordings contextualized by physical space, further optimizing the review process.

Users can interact with the task panel by pressing a button associated with a task to instantly jump to the specific task segments relevant to the current location. This functionality enhances flexibility in reviewing recorded sessions, allowing for targeted review at a specific location without the inefficiencies of linear playback and manual search. The combination of timeline-based navigation and spatial task segment retrieval ensures a more efficient, user-friendly, and structured review workflow.

By integrating automated task segmentation with an intuitive review interface, the system optimizes the review of AR task recording, streamlining task assessment, and improving the overall effectiveness of AR-based training and task guidance.

4 PRELIMINARY FINDINGS FROM USER EVALUATION

To evaluate the effectiveness of the proposed system in improving the efficiency of reviewing AR task recordings compared to the conventional timeline interface, we conducted a user study with 19 participants comparing the proposed Location-based non-linear review interface (see Figure 1(d)) to the baseline of timeline based linear review interface (see Figure 1(c)). Participants were asked to review a 5-minute pre-recorded AR task recording in these two conditions. The recording depicted a task in which cardboard blocks were placed on shelves at various locations (see Figure 1(a)), and

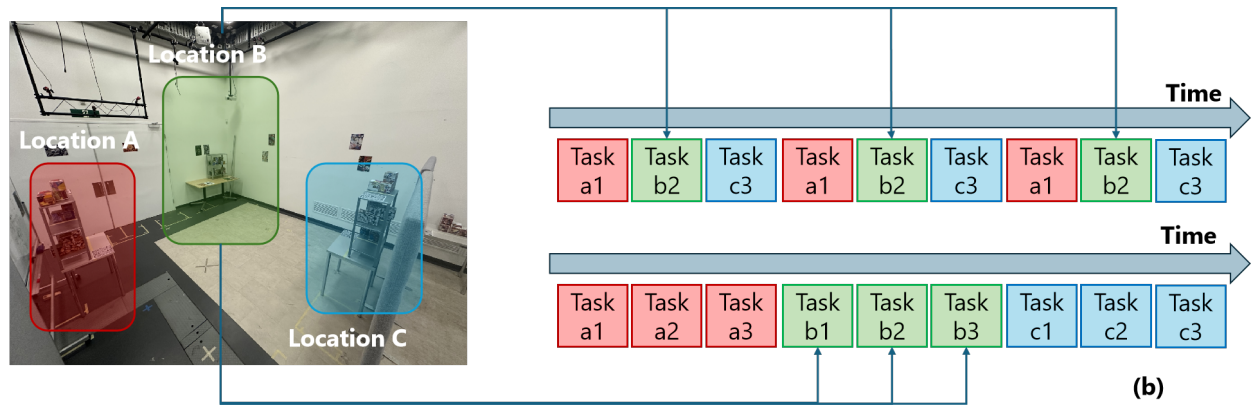


Figure 2: Location-based segmentation and review of AR task recording. An example of a user's review sequence, demonstrating how the system enables users to efficiently navigate task recordings by spatial location rather than linear time, supporting location-aware interactive playback.

participants needed to verify whether the placement order was correct by reviewing the AR recording (see Figure 1(b)). Here we report on preliminary findings based on the participants' feedback.

4.1 User Feedback

Overall, majority of the participants (17 out of 19; 89.5%) selected the Location-based condition as their preferred review method. Participants indicated that the Location-based system was preferred for maintaining focus (16 participants; 84.2%), for reducing mental demand (15 participants; 78.9%), and for reducing physical demand (17 participants; 89.5%). This is also reflected in qualitative feedback. For instance, participants appreciated the ability to “focus on just a single location” and valued that they “don't need to move a lot to watch the video,” which likely contributed to the reduced physical and cognitive load observed.

The location-based timeline and visual cues were also positively received. A participant noted that “the timeline showing which tasks in the location was best,” while another emphasized that they “liked the location-based system... easier to follow without walking around too much.” Additionally, efficient navigation features, such as the ability to “quickly find the right position in the video,” further enhanced usability and task engagement. These qualitative insights underscore the effectiveness of location-anchored and context-aware AR guidance systems in reducing user effort while maintaining task focus.

Participants also indicated that the system was useful and easy to use. A participant mentioned, “This condition is very useful for the expert [reviewing the recording]. Because I can jump to the step that I want to see first or again, so I can check which block is positioned correctly or incorrectly. The user interface is quite easy to use. Therefore, people who don't have any experience with AR can use this system easily.”

Participants also noticed improvements in task efficiency. A participant mentioned, “This condition allowed me to easily fast-forward to the important parts of the recording through the UI and therefore allowed me to easily skip the less important parts while focusing on what mattered,” which highlights the efficiency of non-linear access. The improvements in task efficiency under the Location-based condition can be explained by the benefit of leveraging spatial context for non-linear access to recorded task segments. Unlike the conventional linear playback interface that requires reviewers to scroll or scrub through the entire timeline, location-based segmentation allowed participants to directly access relevant segments based on their physical location or interest. This direct access reduced unnecessary navigation time and minimized

the effort required to locate specific task moments, leading to significantly faster task completion and shorter walking distances.

4.2 Implications

The preliminary findings have important implications for the design of AR interfaces, particularly in the context of task review and training systems.

Supporting Location-based Non-linear Workflows in AR Task reviews can improve user preference. Enabling users to access task-related content non-linearly, based on their immediate needs or physical location, enhances the usability and efficiency of AR systems. This approach is critical for managing complex tasks where time efficiency and rapid access to relevant information are paramount.

Integrating location-based indexing mechanisms as a core feature in AR interfaces can help mirror the physical context of tasks, significantly reducing the task load required to navigate through contextual content.

The positive response to automated task segmentation based on predefined spatial and interaction thresholds suggests that AR systems should incorporate intelligent, context-sensitive segmentation. This reduces reliance on manual navigation and enhances the system's ability to present information in a user-centric manner.

4.3 Limitation and Future Work

While the preliminary findings are promising, several limitations must be acknowledged, particularly highlighted by participant feedback, which suggests areas for further investigation:

Interface improvement: Some users showed resistance to mid-air interactions, favoring more integrated solutions like wall-embedded or handheld UI elements, indicating that the current method may not be the most intuitive or comfortable. Others noted the distraction caused by excessive virtual elements, such as overlapping virtual hands and boxes. These points highlight the need for more ergonomic interaction options and a cleaner visual design to support better user engagement and task clarity.

Fixed Segmentation Criteria: The segmentation criteria used in this study were predetermined (e.g., location radius and dwell time). While effective within the scope of this experiment, they may not be optimal for all use cases or task types. Adapting these criteria based on user feedback and task characteristics could lead to more personalized and effective segmentation strategies.

Unverified Trainee Experience: Although the system was mainly tested for experts reviewing AR task recordings, it is also designed to support trainees navigating unfamiliar environments by

allowing them to efficiently follow a recorded expert's task guidance using spatially segmented task replays. This supports asynchronous collaboration between the trainee and the expert. However, this potential use case was not examined in the current study. Future work should explore how trainees interact with and benefit from location-based playback in actual learning or operational contexts, as this functionality could significantly enhance task comprehension and training efficiency.

Addressing these limitations in future studies could involve expanding test environments, diversifying the participant pool, and iterating on the design based on user feedback to develop more adaptable and user-friendly AR interfaces.

5 CONCLUSION

This project investigated the effectiveness of location-based segmentation and playback of AR task recording in enhancing the efficiency of AR task review systems and improving user experience. As a proof of concept, we presented the design and implementation details of a prototype system which captures AR task recording, segments the recording based on user behavior, and enables intuitive non-linear navigation of the segmented content through location-based playback.

Preliminary findings from a user evaluation underscore the potential of location-based segmentation and playback of AR task recordings, not only to streamline task review processes but also to improve user engagement and satisfaction by aligning with human cognitive and spatial processing. Future work should address the limitations noted regarding interaction modalities and visual clutter to refine the system's applicability in diverse uncontrolled real-world settings.

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